Koneru Lakshmaiah Education Foundation
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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	P.SwethaNagasri
Student Name	Kallu shresta priya
Roll Number	2520030127
Branch	CSE
Course Outcome	Implemented Dynamic Programming techniques to compute Edit Distance efficiently for spell-checking and autocorrect applications.

Case Study Topic:

Spell-checker / Autocorrect – Finding the Closest Dictionary Word Using Edit Distance

Work Done Summary:
Developed a Spell Checker and Autocorrect system using the **Edit Distance (Levenshtein Distance)** algorithm. The system compares a misspelled word with a dictionary word and determines the minimum number of operations required to transform one word into another. The operations considered are **Insertion, Deletion, and Substitution**. Dynamic Programming was used to efficiently compute the edit distance and suggest the closest matching word.

Implementation Details:

- **Language Used:** Java
- **Data Structure Used:** 2D Array (Dynamic Programming Table)
- **Algorithm Used:** Edit Distance (Levenshtein Distance)
- Constructed a DP matrix of size `(m+1) × (n+1)`.
- Initialized boundary conditions for empty strings.
- Compared characters of source and target words.
- If characters matched, copied the diagonal value.
- Otherwise calculated:
  - Insert operation
  - Delete operation
  - Substitute operation
- Stored the minimum value plus one.
- Final edit distance obtained from `dp[m][n]`.

Result: Screenshots / Output

```
PS C:\Users\kallu shresta priya> java EditDistance
Edit Distance = 3
PS C:\Users\kallu shresta priya>
```

The Spell Checker successfully calculated the minimum edit distance between words and identified the closest dictionary match. Dynamic Programming reduced redundant computations and provided an efficient solution suitable for autocorrect systems, search engines, and text-processing applications.

GitHub Link: https://github.com/kallushrestapriya-byte/DSA-case-studies

Student Signature	Faculty Signature
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